

Introduction to Online Convex Optimization

Reading Seminar — Sections 5.5–5.8

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1. Randomized Regularization

1.1 Follow-the-Perturbed-Leader

Algorithm 16 — Follow-the-perturbed-leader for convex losses

- 1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
- 2: Let $x_1 = \mathbb{E}_{n \sim \mathcal{D}}[\arg \min_{x \in \mathcal{K}} \{n^\top x\}]$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t .
- 5: Observe f_t , suffer $f_t(x_t)$, and set $\nabla_t = \nabla f_t(x_t)$.
- 6: Update

$$x_{t+1} = \mathbb{E}_{n \sim \mathcal{D}} \left[\arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^t \nabla_s^\top x + n^\top x \right\} \right] \quad (5.4)$$

7: **end for**

- Resembles a **special case of FTRL** (Follow-the-Regularized-Leader).
- **Key difference:** the regularizer is **randomized** — noise $n \sim \mathcal{D}$ is injected into the objective in place of a fixed regularizer.

1.2 Stability of the Perturbation Distribution

We call a perturbation distribution (σ, L) -**stable** when it is small enough to keep the fake loss controlled, and smooth enough that shifting the noise does not change the leader too much.

Definition $((\sigma, L)$ -stable distribution)

With respect to the norm $\|\cdot\|_a$, a distribution $\mathcal{D}$ over $\mathbb{R}^n$ is (σ, L) -stable if

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u \in \mathbb{R}^n : \int_n |\mathcal{D}(n) - \mathcal{D}(n - u)| \, dn \leq L \|u\|_a^*.$$

The two constants play different roles in the regret bound:

$$\text{Regret}_T \leq \eta D(G^*)^2 LT + \frac{1}{\eta} \sigma D.$$

1.3 Exercise 5.8.6: Uniform Noise Check

The exercise asks us to prove, for $\mathcal{D} = \text{Unif}([0, 1]^n)$ and the Euclidean norm,

$$\sigma_2 < \sqrt{n}, \quad L_2 \leq 1.$$

My reading: the σ bound is fine, but the $L_2 \leq 1$ claim seems too strong for the stability definition above.

Already in one dimension, with $\mathcal{D}(x) = 1_{[0,1]}(x)$ and $0 < u < 1$,

$$\begin{aligned} \int_{\mathbb{R}} |\mathcal{D}(x) - \mathcal{D}(x - u)| \, dx &= |[0, 1] \Delta [u, 1 + u]| \\ &= 2u > u = \|u\|_2. \end{aligned}$$

Thus $L_2 \leq 1$ would require $2u \leq u$, impossible for $u > 0$.

1.4 Uniform Noise: A Safer Stability Bound

Let $N \sim \text{Unif}([0, 1]^n)$. The size term is fine:

$$\sigma_2 = \mathbb{E}[\|N\|_2] \leq \sqrt{\mathbb{E}[\|N\|_2^2]} = \sqrt{\sum_{i=1}^n \mathbb{E}[N_i^2]} = \sqrt{\frac{n}{3}} < \sqrt{n}.$$

For the density-shift term,

$$\begin{aligned} \int |\mathcal{D}(x) - \mathcal{D}(x - u)| \, dx &= |[0, 1]^n \Delta([0, 1]^n + u)| \\ &\leq 2 \sum_{i=1}^n \min(|u_i|, 1) \\ &\leq 2\|u\|_1 \leq 2\sqrt{n}\|u\|_2. \end{aligned}$$

Therefore the usable bounds are $\sigma_2 < \sqrt{n}$ and $L_2 \leq 2\sqrt{n}$. The factor $2\sqrt{n}$ is sharp: take $u = \varepsilon(1, \dots, 1)$ and let $\varepsilon \rightarrow 0^+$.

1.5 Regret Bound for FPL

Theorem (5.8)

Let $\mathcal{D}$ be (σ, L) -stable with respect to the norm $\|\cdot\|_a$. Then FPL (Algorithm 16) attains

$$\text{Regret}_T \leq \eta D(G^*)^2 LT + \frac{1}{\eta} \sigma D,$$

and, optimizing over η ,

$$\text{Regret}_T \leq 2DG^* \sqrt{\sigma LT}.$$

Definition $((\sigma, L)$ -stability)

For $\sigma, L \in \mathbb{R}$, a distribution $\mathcal{D}$ is (σ, L) -stable with respect to $\|\cdot\|_a$ if

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_a^*] = \sigma \quad \text{and} \quad \forall u : \int_n |\mathcal{D}(n) - \mathcal{D}(n - u)| \, dn \leq L \|u\|_a^*.$$

Stability cost. The first term controls how far consecutive expected leaders can move.

Perturbation cost. The second term pays for comparing the fake initial leader with x^* .

1.6 Tool: FTL-BTL Lemma

The proof repeatedly uses the FTL-BTL lemma from Section 5.2.

Lemma (5.4)

If $x_{t+1} \in \arg \min_{x \in \mathcal{K}} \sum_{s=0}^t g_s(x)$, then for every $u \in \mathcal{K}$,

$$\sum_{t=0}^T g_t(u) \geq \sum_{t=0}^T g_t(x_{t+1}).$$

In FPL, the perturbation is treated as a **fake initial loss**

$$g_0^n(x) = \frac{1}{\eta} n^\top x$$

This makes perturbed-leader analysis look exactly like the FTRL proof.

1.7 Proof of Theorem 5.8: Random Leaders

For fixed perturbation n , define the random leader and fake loss:

$$x_t^n = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} \nabla_s^\top x + n^\top x \right\},$$
$$g_0^n(x) := \frac{1}{\eta} n^\top x, \quad g_t(x) := \nabla_t^\top x.$$

Applying Lemma 5.4 to $g_0^n, g_1, \dots, g_T$:

$$\begin{aligned} \mathbb{E}_n \left[\sum_{t=0}^T g_t(u) \right] &\geq \mathbb{E}_n \left[g_0^n(x_1^n) + \sum_{t=1}^T g_t(x_{t+1}^n) \right] \\ &\geq \mathbb{E}_n[g_0^n(x_1^n)] + \sum_{t=1}^T g_t(\mathbb{E}_n[x_{t+1}^n]) \\ &= \mathbb{E}_n[g_0^n(x_1^n)] + \sum_{t=1}^T g_t(x_{t+1}). \end{aligned}$$

1.8 Proof of Theorem 5.8: Regret Decomposition

Taking $u = x^*$ in the previous slide,

$$\begin{aligned}\sum_{t=1}^T \nabla_t^\top (x_t - x^*) &\leq \sum_{t=1}^T g_t(x_t) - \sum_{t=1}^T g_t(x_{t+1}) + \mathbb{E}_n[g_0^n(x^*) - g_0^n(x_1^n)] \\ &\leq \sum_{t=1}^T \nabla_t^\top (x_t - x_{t+1}) + \frac{1}{\eta} \mathbb{E}_n[\|n\|^* \|x^* - x_1^n\|] \\ &\leq \sum_{t=1}^T \nabla_t^\top (x_t - x_{t+1}) + \frac{1}{\eta} \sigma D.\end{aligned}$$

Convexity of f_t and Cauchy-Schwarz give

$$\sum_t f_t(x_t) - \sum_t f_t(x^*) \leq G^* \sum_{t=1}^T \|x_t - x_{t+1}\| + \frac{1}{\eta} \sigma D. \quad (5.5)$$

1.9 Proof of Theorem 5.8: Density Shift

It remains to show that consecutive expected leaders are close. Define

$$h_t(n) = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^{t-1} \nabla_s^\top x + n^\top x \right\}.$$

Then $x_t = \mathbb{E}_{n[h_t(n)]}$. The next leader can be written using the same h_t but with a shifted perturbation:

$$\begin{aligned} x_t &= \int_n h_t(n) \mathcal{D}(n) \, \mathrm{d}n, \\ x_{t+1} &= \int_n h_t(n + \eta \nabla_t) \mathcal{D}(n) \, \mathrm{d}n \\ &= \int_n h_t(n) \mathcal{D}(n - \eta \nabla_t) \, \mathrm{d}n. \end{aligned}$$

Thus stability of decisions reduces to sensitivity of the density $\mathcal{D}$.

1.10 Proof of Theorem 5.8: Bounding the Shift

Using the density-shift expression,

$$\begin{aligned}\|x_t - x_{t+1}\| &= \left\| \int_n h_t(n) \cdot (\mathcal{D}(n) - \mathcal{D}(n - \eta \nabla_t)) \, \mathrm{d}n \right\| \\ &= \left\| \int_n (h_t(n) - h_t(0)) \cdot (\mathcal{D}(n) - \mathcal{D}(n - \eta \nabla_t)) \, \mathrm{d}n \right\| \\ &\leq \int_n \|h_t(n) - h_t(0)\| |\mathcal{D}(n) - \mathcal{D}(n - \eta \nabla_t)| \, \mathrm{d}n \\ &\leq D \int_n |\mathcal{D}(n) - \mathcal{D}(n - \eta \nabla_t)| \, \mathrm{d}n \\ &\leq DL\eta \|\nabla_t\|^* \leq \eta DLG^*.\end{aligned}$$

The only real input here is (σ, L) -stability with $u = \eta \nabla_t$.

1.11 Proof of Theorem 5.8: Final Bound

Substitute the shift bound into (5.5):

$$\begin{aligned}\text{Regret}_T &\leq G^* \sum_{t=1}^T \|x_t - x_{t+1}\| + \frac{1}{\eta} \sigma D \\ &\leq G^* \sum_{t=1}^T \eta D L G^* + \frac{1}{\eta} \sigma D \\ &= \eta D L (G^*)^2 T + \frac{1}{\eta} \sigma D.\end{aligned}$$

Optimizing at $\eta = \sqrt{\frac{\sigma}{L(G^*)^2 T}}$ gives the stated theorem bound:

$$\text{Regret}_T \leq 2 D G^* \sqrt{\sigma L T}.$$

With the corrected uniform-cube bound $L \leq 2\sqrt{n}$ and $\sigma \leq \sqrt{n}$, this gives a loose $O(DG\sqrt{nT})$ bound.

1.12 FPL for Linear Losses

Algorithm 17 — FPL for linear losses

- 1: Input: $\eta > 0$, distribution $\mathcal{D}$ over $\mathbb{R}^n$, decision set $\mathcal{K} \subseteq \mathbb{R}^n$.
- 2: Sample $n_0 \sim \mathcal{D}$. Let $\hat{x}_1 \in \arg \min_{x \in \mathcal{K}} \{-n_0^\top x\}$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict $\hat{x}_t$.
- 5: Observe the linear loss function, suffer loss $g_t^\top \hat{x}_t$.
- 6: Update

$$\hat{x}_{t+1} = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^t g_s^\top x + n_0^\top x \right\}$$

7: **end for**

Linear loss. The exact expectation in Algorithm 16 can be replaced by one perturbation sample.

Expected regret. The guarantee is taken over the sampled perturbation.

1.13 Why One Sample is Enough for Linear Losses

Let

$$w_t(n) = \arg \min_{x \in \mathcal{K}} \left\{ \eta \sum_{s=1}^t g_s^\top x + n^\top x \right\}.$$

Algorithm 16 predicts $x_t = \mathbb{E}_n[w_{t-1}(n)]$.

For a linear loss $f_t(x) = g_t^\top x$,

$$f_t(x_t) = f_t(\mathbb{E}_n[w_{t-1}(n)]) = \mathbb{E}_n[f_t(w_{t-1}(n))].$$

Therefore sampling one $n_0 \sim \mathcal{D}$ and playing $\hat{x}_t = w_{t-1}(n_0)$ has the same expected loss as the exact expectation-based prediction.

1.14 Corollary 5.9: Expected Regret for Linear FPL

Theorem (Corollary 5.9)

For Algorithm 17,

$$\mathbb{E}_{n_0 \sim \mathcal{D}} \left[\sum_{t=1}^T f_t(\hat{x}_t) - \sum_{t=1}^T f_t(x^*) \right] \leq \eta L D (G^*)^2 T + \frac{1}{\eta} \sigma D.$$

Computational point. Algorithm 17 only needs a linear optimization oracle over $\mathcal{K}$; the set need not even be convex for the oracle call itself.

1.15 FPL for Prediction from Expert Advice

Algorithm 18 – FPL for prediction from expert advice

- 1: Input: $\eta > 0$.
- 2: Draw n exponentially distributed variables $n(i) \sim e^{-\eta x}$.
- 3: Let $x_1 = \arg \min_{e_i \in \Delta_n} \{-e_i^\top n\}$.
- 4: **for** $t = 1$ to T **do**
- 5: Predict using expert i_t such that $\hat{x}_t = e_{i_t}$.
- 6: Observe the loss vector, suffer loss $g_t^\top \hat{x}_t = g_t(i_t)$.
- 7: Update (w.l.o.g. choose $\hat{x}_{t+1}$ to be a vertex)

$$\hat{x}_{t+1} = \arg \min_{x \in \Delta_n} \left\{ \sum_{s=1}^t g_s^\top x - n^\top x \right\}$$

8: **end for**

- Special case of Algorithm 17: the simplex Δ_n , costs in $[0, 1]$, and one-sided exponential noise $n(i) \sim e^{-\eta x}$ – the first use of perturbation in decision making (Hannan [1957]).
- The generic FPL bound (Corollary 5.9) is loose here, so Theorem 5.10 gives a dedicated analysis, tight up to constants.

1.16 Exponential Noise for Expert Advice

For expert advice, the perturbation is one-sided exponential noise:

$$\Pr[n(i) \leq x] = 1 - e^{-\eta x} \quad (x \geq 0).$$

The update chooses the expert with smallest cumulative loss after subtracting its random bonus:

$$\hat{x}_{t+1} = \arg \min_{x \in \Delta_n} \left\{ \sum_{s=1}^t g_s^\top x - n^\top x \right\}.$$

Corollary 5.9 is too loose for this simplex/exponential-noise case, so Theorem 5.10 uses a direct leader-change analysis.

1.17 Regret Bound for Expert-Advice FPL

Theorem (5.10)

Algorithm 18 outputs predictions $\hat{x}_1, \dots, \hat{x}_T \in \Delta_n$ with

$$(1 - \eta) \mathbb{E} \left[\sum_t g_t^\top \hat{x}_t \right] \leq \min_{x^* \in \Delta_n} \sum_t g_t^\top x^* + \frac{4 \log n}{\eta}.$$

Optimizing η . Choosing $\eta = \sqrt{(\log n)/T}$ gives

$$\text{Regret}_T = O\left(\sqrt{T \log n}\right),$$

matching the **Hedge** guarantee (Theorem 1.5) up to constants.

Why a separate analysis? — the order gap.

- **Generic FPL (Cor. 5.9)** on Δ_n : the corrected uniform-cube bound has $\sigma \leq \sqrt{n}$, $L \leq 2\sqrt{n}$, giving $\text{Regret}_T = O\left(DG\sqrt{nT}\right)$ — a polynomial-in- n loss versus the $O\left(\sqrt{T}\right)$ of OGD (Theorem 3.1).
- **Expert-advice FPL (Thm 5.10)**, with exponential noise, removes that polynomial-in- n penalty: only a $\sqrt{\log n}$ dependence remains, on par with the standard multiplicative-weights / Hedge bound $O\left(\sqrt{T \log n}\right)$.

1.18 Proof of Theorem 5.10 (1/2)

Proof. Set $g_0 = -n$. Applying **Lemma 5.4** (FTL-BTL) to $f_t(x) = g_t^\top x$,

$$\mathbb{E} \left[\sum_{t=0}^T g_t^\top u \right] \geq \mathbb{E} \left[\sum_{t=0}^T g_t^\top \hat{x}_{t+1} \right].$$

Therefore, telescoping and isolating the $t = 0$ term,

$$\begin{aligned} \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^\star) \right] &\leq \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \right] + \mathbb{E}[g_0^\top (x^\star - x_1)] \\ &\leq \mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \right] + \mathbb{E}[\|n\|_\infty \|x^\star - x_1\|_1] \\ &\leq \sum_{t=1}^T \mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] + \frac{4}{\eta} \log n. \quad (5.6) \end{aligned}$$

The 2nd line uses the **generalized Cauchy–Schwarz** inequality; the 3rd uses

$$\mathbb{E}_{n \sim \mathcal{D}}[\|n\|_\infty] \leq \frac{2 \log n}{\eta} \quad (\text{exercise}), \quad \|x^\star - x_1\|_1 \leq 2.$$

1.19 Exercise Bound for $\|n\|_\infty$: A Careful Version

Let $N_i \sim \text{Exp}(\eta)$ independently and $M = \|N\|_\infty = \max_i N_i$.

My reading: the exercise bound $\mathbb{E}[M] \leq \frac{2 \log n}{\eta}$ needs a small qualification as stated.

One safe upper bound. By Boole / union bound,

Why a qualification is needed: $n = 2$.

$$\begin{aligned}\mathbb{E}[M] &= \int_0^\infty \Pr[M \geq t] dt \\ &= \int_0^\infty (1 - (1 - e^{-\eta t})^2) dt \\ &= \frac{3}{2\eta} > \frac{2 \log 2}{\eta}.\end{aligned}$$

$$\begin{aligned}\Pr[M \geq t] &\leq \sum_{i=1}^n \Pr[N_i \geq t] \\ &= ne^{-\eta t}.\end{aligned}$$

Also $\Pr[M \geq t] \leq 1$. With $a = \frac{\log n}{\eta}$,

$$\begin{aligned}\mathbb{E}[M] &\leq a + \int_a^\infty ne^{-\eta t} dt \\ &= \frac{1 + \log n}{\eta}.\end{aligned}$$

Thus we can use $\mathbb{E}[\|N\|_\infty] \leq \frac{1 + \log n}{\eta}$. For $n \geq 3$ this implies the stated $\frac{2 \log n}{\eta}$ bound.

1.20 Proof of Theorem 5.10 (2/2)

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \leq 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \leq \|g_t\|_\infty \cdot \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] \leq \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned} \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] &= 1 - \Pr[n(i_t) > v + g_t(i_t) \mid n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x} dx}{\int_v^{\infty} \eta e^{-\eta x} dx} = 1 - e^{-\eta g_t(i_t)} \leq \eta g_t(i_t) = \eta g_t^\top \hat{x}_t. \end{aligned}$$

Substituting into (5.6),

$$\mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^*) \right] \leq \eta \sum_t \mathbb{E}_t [g_t^\top \hat{x}_t] + \frac{4 \log n}{\eta},$$

which rearranges to the theorem. □

1.20 Proof of Theorem 5.10 (2/2)

Per-step term. Bounding by the probability that the leader changes, times $\|g_t\|_\infty \leq 1$ (losses bounded by one):

$$\mathbb{E}[g_t^\top (\hat{x}_t - \hat{x}_{t+1}) \mid \hat{x}_t] \leq \|g_t\|_\infty \cdot \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] \leq \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t].$$

$\hat{x}_t = e_{i_t}$ leads iff $n(i_t) > v$ for some v fixed by the history; it **stays** the leader iff $n(i_t) > v + g_t(i_t)$. By the **memorylessness** of the exponential distribution,

$$\begin{aligned} \Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t] &\leq 1 - \Pr[n(i_t) > v + g_t(i_t) \mid n(i_t) > v] \\ &= 1 - \frac{\int_{v+g_t(i_t)}^{\infty} \eta e^{-\eta x} dx}{\int_v^{\infty} \eta e^{-\eta x} dx} = 1 - e^{-\eta g_t(i_t)} \leq \eta g_t(i_t) = \eta g_t^\top \hat{x}_t. \end{aligned}$$

My reading: I use “ $\leq$ ” for this step; the next slide explains why.

Substituting into (5.6),

$$\mathbb{E} \left[\sum_{t=1}^T g_t^\top (\hat{x}_t - x^*) \right] \leq \eta \sum_t \mathbb{E}_t [g_t^\top \hat{x}_t] + \frac{4 \log n}{\eta},$$

which rearranges to the theorem. □

1.21 The Marked Equality is Only “ $\leq$ ”: an $N = 2$ Check

The proof’s event is a **sufficient** condition for staying leader, not a necessary one: it ignores that competitors also pay losses.

Expert	Before g_t	After $g_{t(1)} = g_{t(2)} = 1$
1 = leader	S_1	$S_1 + 1$
2	$S_2 > S_1$	$S_2 + 1 > S_1 + 1$

Equal losses shift both scores by the same amount, so the arg min cannot change:

$$\Pr[\hat{x}_t \neq \hat{x}_{t+1} \mid \hat{x}_t = e_1] = 0.$$

But the displayed equality would give

$$1 - e^{-\eta g_{t(1)}} = 1 - e^{-\eta} > 0.$$

Thus the marked step should be “ $\leq$ ”, not “ $=$ ”. The direction is still enough for the regret bound, so Theorem 5.10 is unaffected.

2. Adaptive Gradient Descent (AdaGrad)

2.1 Learning the Regularizer Online

Motivation. RFTL's regret bound (5.7) depends on the regularizer R ; the optimal R depends on both $\mathcal{K}$ and the cost sequence. So we learn it online — AdaGrad optimizes the regularization (line 5) to shrink the dominant gradient-norm term in (5.7).

Algorithm 19 — AdaGrad

- 1: Input: parameters $\eta, x_1 \in \mathcal{K}$.
- 2: Initialize: $G_0 = 0$.
- 3: **for** $t = 1$ to T **do**
- 4: Predict x_t , suffer loss $f_t(x_t)$.
- 5: Update $G_t = G_{t-1} + \nabla_t \nabla_t^\top$ and define

$$\text{(diagonal)} \quad H_t = \arg \min_{H \succeq 0, H = \text{diag}(H)} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = (\text{diag}(G_t))^{\frac{1}{2}}$$

$$\text{(full matrix)} \quad H_t = \arg \min_{H \succeq 0} \{G_t \cdot H^{-1} + \text{Tr}(H)\} = G_t^{\frac{1}{2}}$$

- 6: Update $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t$, $x_{t+1} = \arg \min_{x \in \mathcal{K}} \|y_{t+1} - x\|_{H_t}^2$.
 - 7: **end for**
-

2.2 What Line 5 is Optimizing

AdaGrad chooses the matrix H_t that would have made the **observed** gradients small, with a trace penalty to keep the metric from blowing up:

$$H_t = \arg \min_H \{G_t \cdot H^{-1} + \text{Tr}(H)\}, \quad G_t = \sum_{s=1}^t \nabla_s \nabla_s^\top.$$

Definition (Two comparator classes)

$$\mathcal{H}_1 = \{H : H = \text{diag}(H), H \succeq 0, \text{Tr}(H) \leq 1\}, \quad \mathcal{H}_2 = \{H : H \succeq 0, \text{Tr}(H) \leq 1\}.$$

Lemma (5.11)

For $i \in \{1, 2\}$, with the corresponding final AdaGrad matrix H_T ,

$$\sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^T \|\nabla_t\|_H^{*2}} = \text{Tr}(H_T).$$

2.3 Regret Bound: Compete with the Best Fixed Metric

Theorem (5.12)

Let x_t be generated by Algorithm 19. With $\eta = \frac{D_\infty}{\sqrt{2}}$ for the diagonal version and $\eta = \frac{D}{\sqrt{2}}$ for the full-matrix version, for any $x^* \in \mathcal{K}$,

$$\text{Regret}_{T(\text{AdaGrad-diag})} \leq \sqrt{2}D_\infty \sqrt{\min_{H \in \mathcal{H}_1} \sum_t \|\nabla_t\|_H^{*2}},$$

$$\text{Regret}_{T(\text{AdaGrad-full})} \leq \sqrt{2}D \sqrt{\min_{H \in \mathcal{H}_2} \sum_t \|\nabla_t\|_H^{*2}}.$$

Reading. AdaGrad is nearly as good as the best fixed quadratic regularizer in hindsight, chosen from the diagonal or full PSD trace ball.

Why the theorem follows from Lemma 5.11.

The adaptive choice makes $\text{Tr}(H_T)$ exactly encode the best hindsight gradient norm over $\mathcal{H}_i$.

Why there is still a price.

The regularizer changes over time, so the proof needs one extra drift term involving $H_t - H_{t-1}$.

2.4 When AdaGrad Beats OGD: Upper Bounds

Decision set.

$$\mathcal{K} = [0, 1]^d.$$

Diameters.

$$D_\infty = \sup_{x, y \in \mathcal{K}} \|x - y\|_\infty = 1,$$

$$D = \sup_{x, y \in \mathcal{K}} \|x - y\|_2 = \sqrt{d}, \quad \|x - y\|_2^2 = \sum_{i=1}^d (x_i - y_i)^2 \leq d.$$

D_∞ is tight at $x = (0, \dots, 0)$, $y = (1, 0, \dots, 0)$; D is tight at $x = (0, \dots, 0)$, $y = (1, \dots, 1)$.

Coordinate energy.

$$a_i := \sum_{t=1}^T g_{t,i}^2, \quad \text{diag}(G_T) = \text{diag}(a_1, \dots, a_d).$$

Theorem (Unit cube upper bounds)

$$\begin{aligned} \text{Regret}_{T(\text{AdaGrad-diag})} &\leq \sqrt{2} D_\infty \text{Tr}\left((\text{diag}(G_T))^{\frac{1}{2}}\right) \\ &= \sqrt{2} \sum_{i=1}^d \sqrt{a_i}, \\ \text{Regret}_{T(\text{OGD})} &\leq D \sqrt{2 \sum_{t=1}^T \|g_t\|_2^2} \\ &= \sqrt{2d \sum_{i=1}^d a_i}. \end{aligned}$$

Takeaway. AdaGrad sees $\sum_i \sqrt{a_i}$, while OGD sees $\sqrt{d \sum_i a_i}$. Thus AdaGrad gains when the gradient energy a_i is concentrated on few coordinates.

2.5 Example: Sparse Gradients

Definition.

$$g_t = (1, 0, \dots, 0).$$

Then

$$a_1 = T, \quad a_2 = \dots = a_d = 0.$$

Therefore

$$\frac{\text{OGD bound}}{\text{AdaGrad bound}} = \frac{\sqrt{2dT}}{\sqrt{2T}} = \sqrt{d}.$$

Takeaway. If the gradients concentrate on one coordinate, the AdaGrad upper bound is a factor $\sqrt{d}$ better than the OGD upper bound.

Theorem (Substitute into the bounds)

$$\text{Regret}_{T(\text{AdaGrad-diag})} \leq \sqrt{2} \sum_{i=1}^d \sqrt{a_i} = \sqrt{2T},$$

$$\text{Regret}_{T(\text{OGD})} \leq \sqrt{2d \sum_{i=1}^d a_i} = \sqrt{2dT}.$$

2.6 Example: Dense Gradients

Definition.

$$g_t = (1, 1, \dots, 1).$$

Then all coordinates have the same energy:

$$a_1 = \dots = a_d = T.$$

Theorem (Substitute into the bounds)

$$\text{Regret}_{T(\text{AdaGrad-diag})} \leq \sqrt{2} \sum_{i=1}^d \sqrt{a_i} = \sqrt{2}d\sqrt{T},$$

$$\text{Regret}_{T(\text{OGD})} \leq \sqrt{2d \sum_{i=1}^d a_i} = \sqrt{2d \cdot dT} = \sqrt{2}d\sqrt{T}.$$

Thus $\text{Regret}_{T(\text{AdaGrad-diag})}$ and $\text{Regret}_{T(\text{OGD})}$ have the same order on dense gradients.

Takeaway. AdaGrad is not always better than OGD. It is strong when the gradient sequence is coordinate-sparse; in dense cases, the two bounds have the same order.

2.7 Analysis Roadmap for Theorem 5.12

The proof splits AdaGrad regret into two terms, then controls each by the same final quantity $\text{Tr}(H_T)$.

Lemma (5.13)

$$\text{Regret}_T \leq \frac{\eta}{2} (G_T \cdot H_T^{-1} + \text{Tr}(H_T)) + \frac{1}{2\eta} \sum_{t=0}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2.$$

Lemma 5.14. Gradient term:

$$G_T \cdot H_T^{-1} \leq \text{Tr}(H_T).$$

Lemma 5.15. Drift term:

$$\sum_t \|x_t - x^*\|_{H_t - H_{t-1}}^2 \leq D_*^2 \text{Tr}(H_T).$$

Choosing η balances the two terms and recovers Theorem 5.12.

2.8 Lemma 5.13: Proof Details

By convexity, it suffices to bound the linearized regret:

$$\text{Regret}_T \leq \sum_{t=1}^T \nabla_t^\top (x_t - x^*).$$

Changing metrics telescope. For

$$S = \sum_{t=1}^T \left(\|x_t - x^*\|_{H_t}^2 - \|x_{t+1} - x^*\|_{H_t}^2 \right).$$

One-step inequality. With $y_{t+1} = x_t - \eta H_t^{-1} \nabla_t$ and $x_{t+1} = \Pi_K^{H_t}(y_{t+1})$,

$$\|y_{t+1} - x^*\|_{H_t}^2 = \|x_t - x^*\|_{H_t}^2 - 2\eta \nabla_t^\top (x_t - x^*) + \eta^2 \nabla_t^\top H_t^{-1} \nabla_t,$$

$$\|x_{t+1} - x^*\|_{H_t}^2 \leq \|y_{t+1} - x^*\|_{H_t}^2,$$

hence

$$\nabla_t^\top (x_t - x^*) \leq \frac{\eta}{2} \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} \left(\|x_t - x^*\|_{H_t}^2 - \|x_{t+1} - x^*\|_{H_t}^2 \right). \quad (*)$$

Sum (*).

$$\sum_{t=1}^T \nabla_t^\top (x_t - x^*) \leq \frac{\eta}{2} \sum_{t=1}^T \nabla_t^\top H_t^{-1} \nabla_t + \frac{1}{2\eta} S.$$

Therefore

$$\text{Regret}_T \leq \frac{\eta}{2} (G_T \cdot H_T^{-1} + \text{Tr}(H_T)) + \frac{1}{2\eta} \sum_{t=0}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2.$$

$$S = \|x_1 - x^*\|_{H_0}^2 - \|x_{T+1} - x^*\|_{H_T}^2$$

$$+ \sum_{t=1}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2.$$

Dropping the final non-positive term gives the drift term.

BTL for the gradient term. Define

$$\Psi_{t(H)} = \nabla_t \nabla_t^\top \cdot H^{-1}, \quad \Psi_0(H) = \text{Tr}(H).$$

Since H_t minimizes $\sum_{i=0}^t \Psi_{i(H)}$,

$$\begin{aligned} \sum_{t=1}^T \nabla_t^\top H_t^{-1} \nabla_t &= \sum_{t=1}^T \Psi_{t(H_t)} \leq \sum_{t=1}^T \Psi_{t(H_T)} + \Psi_0(H_T) - \Psi_0(H_0) \\ &= G_T \cdot H_T^{-1} + \text{Tr}(H_T). \end{aligned}$$

2.9 Proposition 5.16: Two Matrix Optimizations

The optimizer in AdaGrad line 5 comes from the second problem below. For $A \succeq 0$,

$$(P_1) \quad \min_{X \succ 0, \operatorname{Tr}(X) \leq 1} A \cdot X^{-1},$$

$$(P_2) \quad \min_{X \succ 0} \{A \cdot X^{-1} + \operatorname{Tr}(X)\}.$$

Exercise 11. The trace-ball problem (P_1) is exactly the exercise: prove its minimizer and minimum value.

Proposition 5.16 says

$$X_1^* = \frac{A^{\frac{1}{2}}}{\operatorname{Tr}(A^{\frac{1}{2}})}, \quad \min(P_1) = \operatorname{Tr}(A^{\frac{1}{2}})^2,$$

$$X_2^* = A^{\frac{1}{2}}, \quad \min(P_2) = 2 \operatorname{Tr}(A^{\frac{1}{2}}).$$

The diagonal case is the same proof after replacing A by $\operatorname{diag}(A)$. I sketch both facts before using them in Lemma 5.14.

2.10 Proposition 5.16: Trace-Ball Problem (Exercise 11)

Let $S = \text{Tr}(A^{\frac{1}{2}})$. For any $X \succ 0$, Frobenius Cauchy-Schwarz gives

$$\begin{aligned} S &= \text{Tr}(A^{\frac{1}{2}}) = \text{Tr}(A^{\frac{1}{2}} X^{-\frac{1}{2}} X^{\frac{1}{2}}) \\ &\leq \sqrt{\text{Tr}(A^{\frac{1}{2}} X^{-1} A^{\frac{1}{2}})} \sqrt{\text{Tr}(X)} \\ &= \sqrt{A \cdot X^{-1}} \sqrt{\text{Tr}(X)}. \end{aligned}$$

Therefore, if $\text{Tr}(X) \leq 1$,

$$A \cdot X^{-1} \geq S^2 = \text{Tr}(A^{\frac{1}{2}})^2.$$

Equality in Cauchy-Schwarz requires $X^{-\frac{1}{2}} A^{\frac{1}{2}} = c X^{\frac{1}{2}}$, hence $A^{\frac{1}{2}} = c X$. The trace constraint is tight, so

$$X = \frac{A^{\frac{1}{2}}}{\text{Tr}(A^{\frac{1}{2}})}.$$

2.11 Proposition 5.16: Penalized Problem

Use the same inequality, now without the trace constraint. Put $t = \text{Tr}(X) > 0$ and $S = \text{Tr}(A^{\frac{1}{2}})$. Then

$$A \cdot X^{-1} \geq \frac{S^2}{t}.$$

So the objective is bounded by a one-variable problem:

$$\begin{aligned} A \cdot X^{-1} + \text{Tr}(X) &\geq \frac{S^2}{t} + t \\ &= \frac{(t - S)^2}{t} + 2S \geq 2S. \end{aligned}$$

The lower bound is attained at $t = S$ and equality in the Frobenius Cauchy-Schwarz step, which together give

$$X = A^{\frac{1}{2}}.$$

2.12 Proposition 5.16: Diagonal Case

If X is diagonal, only the diagonal of A is visible:

$$A \cdot X^{-1} = \text{diag}(A) \cdot X^{-1}.$$

Thus the same proof applies with $B = \text{diag}(A)$:

$$\text{trace-ball optimizer: } X_1^* = \frac{B^{\frac{1}{2}}}{\text{Tr}(B^{\frac{1}{2}})},$$

$$\text{penalized optimizer: } X_2^* = B^{\frac{1}{2}}.$$

Applying the second formula to $A = G_T$ gives the AdaGrad matrices:

$$H_T = G_T^{\frac{1}{2}} \quad \text{or} \quad H_T = (\text{diag}(G_T))^{\frac{1}{2}}.$$

2.13 Lemma 5.14: Gradient Term is at Most Trace

For the full-matrix version,

$$G_T \cdot H_T^{-1} = G_T \cdot G_T^{-\frac{1}{2}} = \text{Tr}\left(G_T^{\frac{1}{2}}\right) = \text{Tr}(H_T).$$

For the diagonal version, H_T ignores off-diagonal entries:

$$G_T \cdot H_T^{-1} = \text{diag}(G_T) \cdot (\text{diag}(G_T))^{-\frac{1}{2}} = \text{Tr}\left((\text{diag}(G_T))^{\frac{1}{2}}\right) = \text{Tr}(H_T).$$

Therefore, in both cases,

$$G_T \cdot H_T^{-1} \leq \text{Tr}(H_T).$$

2.14 Lemma 5.15: Diagonal Drift Term

For diagonal AdaGrad, $G_t \succeq G_{t-1}$ implies $H_t - H_{t-1} \succeq 0$. Since for diagonal H , $x^\top H x \leq \|x\|_\infty^2 \text{Tr}(H)$,

$$\begin{aligned} \sum_{t=1}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2 &\leq \sum_{t=1}^T D_\infty^2 \text{Tr}(H_t - H_{t-1}) \\ &= D_\infty^2 \sum_{t=1}^T (\text{Tr}(H_t) - \text{Tr}(H_{t-1})) \\ &\leq D_\infty^2 \text{Tr}(H_T). \end{aligned}$$

The trace telescopes because the metric increments are PSD.

2.15 Lemma 5.15: Full-Matrix Drift Term

In the full-matrix case, $H_t - H_{t-1} \succeq 0$ and $\lambda_{\max}(A) \leq \text{Tr}(A)$ for $A \succeq 0$:

$$\begin{aligned} \sum_{t=1}^T \|x_t - x^*\|_{H_t - H_{t-1}}^2 &\leq \sum_{t=1}^T D^2 \lambda_{\max}(H_t - H_{t-1}) \\ &\leq D^2 \sum_{t=1}^T \text{Tr}(H_t - H_{t-1}) \\ &= D^2 \sum_{t=1}^T (\text{Tr}(H_t) - \text{Tr}(H_{t-1})) \\ &\leq D^2 \text{Tr}(H_T). \end{aligned}$$

This proves Lemma 5.15.

2.16 Putting the Pieces Together

Lemma 5.13 plus Lemmas 5.14 and 5.15 gives

$$\begin{aligned}\text{Regret}_T &\leq \frac{\eta}{2}(2 \text{Tr}(H_T)) + \frac{D_*^2}{2\eta} \text{Tr}(H_T) \\ &= \left(\eta + \frac{D_*^2}{2\eta} \right) \text{Tr}(H_T),\end{aligned}$$

where $D_* = D_\infty$ for diagonal AdaGrad and $D_* = D$ for full AdaGrad.

Choosing $\eta = \frac{D_*}{\sqrt{2}}$ yields

$$\text{Regret}_T \leq \sqrt{2}D_* \text{Tr}(H_T).$$

Finally, Lemma 5.11 converts $\text{Tr}(H_T)$ back to the best hindsight metric:

$$\text{Tr}(H_T) = \sqrt{\min_{H \in \mathcal{H}_i} \sum_{t=1}^T \|\nabla_t\|_H^{*2}},$$

which is exactly Theorem 5.12.

3. Bibliographic Remarks

3.1 Notes & References

- **Origins.** Regularization for online learning: Grove et al. [2001]; Kivinen & Warmuth [2001].
- **Follow-the-(Perturbed-)Leader.** Kalai & Vempala [2005] coined *follow-the-leader* and analyzed random perturbation as a regularizer (FPL), reviving an idea of Hannan [1957].
- **Follow-the-Regularized-Leader.** *FTRL* coined by Shalev-Shwartz & Singer [2007]; identical *RFTL* by Abernethy et al. [2008]. Hazan & Kale [2008]: *RFTL* $\equiv$ Online Mirror Descent.
- **AdaGrad.** Duchi et al. [2010, 2011]; diagonal version in parallel by McMahan & Streeter [2010]. Analysis here follows Gupta et al. [2017].
- **Adaptive regularization in deep learning.** AdaGrad, RMSprop [Tieleman & Hinton 2012], Adam [Kingma & Ba 2014]; survey: Goodfellow et al. [2016].
- **Randomization $\leftrightarrow$ regularization.** In special cases, adding randomization $\approx$ deterministic strongly convex regularization [Abernethy et al. 2014, 2016].

4. Exercises



4.1 Exercise 1(a): Dual of a Matrix Norm

Claim. Let $A \succ 0$ and define $\|x\|_A := \sqrt{x^\top A x}$. Then its dual norm is

$$\|y\|_A^* = \|y\|_{A^{-1}} = \sqrt{y^\top A^{-1} y}.$$

Proof. By definition,

$$\|y\|_A^* = \sup_{\|x\|_A \leq 1} x^\top y.$$

Put $z = A^{\frac{1}{2}}x$, so $x = A^{-\frac{1}{2}}z$ and $\|x\|_A = \|z\|_2$. Hence

$$\begin{aligned} \|y\|_A^* &= \sup_{\|z\|_2 \leq 1} z^\top A^{-\frac{1}{2}} y \\ &= \|A^{-\frac{1}{2}} y\|_2 = \sqrt{y^\top A^{-1} y}. \end{aligned}$$

The supremum is attained in the direction $z = A^{-\frac{1}{2}}y / \|A^{-\frac{1}{2}}y\|_2$ when $y \neq 0$. □

4.2 Exercise 1(b): Generalized Cauchy–Schwarz

Claim. For any norm $\|\cdot\|$ with dual norm $\|y\|_* := \sup_{\|z\| \leq 1} z^\top y$,

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

Proof. If $x = 0$ (or $y = 0$), both sides equal 0 and the inequality holds. So assume $x \neq 0$ from here on.

Since $x \neq 0$, the unit vector $u = x/\|x\|$ satisfies $\|u\| = 1$, hence it is admissible in the supremum defining the dual norm. Therefore

$$\|y\|_* = \sup_{\|z\| \leq 1} z^\top y \geq u^\top y = \frac{x^\top y}{\|x\|}.$$

Multiplying both sides by $\|x\| > 0$ yields

$$x^\top y \leq \|x\| \cdot \|y\|_*.$$

□

4.3 Problem 5: Negative Entropy Setup

Definition (Setup)

The decision set is the n -dimensional simplex

$$\Delta_n = \left\{ x \in \mathbb{R}_{\geq 0}^n : \sum_{i=1}^n x_i = 1 \right\},$$

and the regularizer is negative entropy

$$R(x) = \sum_{i=1}^n x_i \log x_i, \quad 0 \log 0 := 0.$$

The Bregman divergence is

$$D_{R(x,y)} := R(x) - R(y) - \langle \nabla R(y), x - y \rangle.$$

Since $\nabla R(y)_i = \log y_i + 1$,

$$\langle \nabla R(y), x - y \rangle = \sum_i (\log y_i + 1)(x_i - y_i).$$

4.4 Problem 5: Bregman Divergence is KL

Substitute the previous identities into the Bregman divergence:

$$\begin{aligned} D_{R(x,y)} &= \sum_i x_i \log x_i - \sum_i y_i \log y_i - \sum_i (\log y_i + 1)(x_i - y_i) \\ &= \sum_i x_i \log \left(\frac{x_i}{y_i} \right) - \sum_i x_i + \sum_i y_i. \end{aligned}$$

For $x, y \in \Delta_n$, the last two terms cancel, hence

$$D_{R(x,y)} = \sum_i x_i \log \left(\frac{x_i}{y_i} \right).$$

Theorem (Relative entropy)

$$D_{R(x,y)} = \text{KL}(x \parallel y) = \sum_i x_i \log \left(\frac{x_i}{y_i} \right).$$

This is the relative entropy geometry underlying multiplicative-weights style updates.

4.5 Problem 5: Diameter Bound

For this exercise, the diameter is the range of the regularizer on the simplex:

$$D_R := \max_{x \in \Delta_n} R(x) - \min_{x \in \Delta_n} R(x).$$

Note: if we instead used $\max_{x,y \in \Delta_n} D_{R(x,y)}$, the value would be infinite on the boundary whenever $y_i = 0$ and $x_i > 0$.

Minimum. By convexity, the minimum occurs at the uniform point $u = (\frac{1}{n}, \dots, \frac{1}{n})$:

$$R(u) = \sum_i \frac{1}{n} \log\left(\frac{1}{n}\right) = -\log n.$$

Maximum. The maximum occurs at a vertex, say $e_1 = (1, 0, \dots, 0)$:

$$R(e_1) = 1 \log 1 + \sum_{i \neq 1} 0 \log 0 = 0.$$

Therefore

$$D_R = 0 - (-\log n) = \log n.$$

Theorem (Diameter bound)

$$D_R \leq \log n, \quad \text{in fact} \quad D_R = \log n.$$

4.6 Problem 5: Projection Objective

Let $y \in \mathbb{R}_{>0}^n$ and project it onto the simplex:

$$x^\star = \arg \min_{x \in \Delta_n} D_{R(x,y)}.$$

From the previous calculation,

$$D_{R(x,y)} = \sum_i x_i \log\left(\frac{x_i}{y_i}\right) - \sum_i x_i + \sum_i y_i.$$

Since $x \in \Delta_n$, $\sum_i x_i = 1$, and $\sum_i y_i$ is constant in x . Thus the essential problem is

$$\min_{x \in \Delta_n} \sum_i x_i \log\left(\frac{x_i}{y_i}\right).$$

4.7 Problem 5: Projection First-Order Condition

The Lagrangian for the constraint $\sum_i x_i = 1$ is

$$\mathcal{L}(x, \lambda) = \sum_i x_i \log\left(\frac{x_i}{y_i}\right) + \lambda \left(\sum_i x_i - 1 \right).$$

The first-order condition gives

$$\frac{\partial \mathcal{L}}{\partial x_i} = \log\left(\frac{x_i}{y_i}\right) + 1 + \lambda = 0.$$

Hence $\frac{x_i}{y_i} = C$ for a constant C , so $x_i = Cy_i$.

4.8 Problem 5: Projection Formula

Enforcing $\sum_i x_i = 1$ gives

$$C \sum_i y_i = 1, \quad C = \frac{1}{\|y\|_1}, \quad x_i^* = \frac{y_i}{\|y\|_1}.$$

Theorem (Projection formula)

$$\Pi_{\Delta_n}^R(y) = \frac{y}{\|y\|_1}.$$

Thus Bregman projection for negative entropy simply rescales a positive vector so that its coordinates sum to one.

4.9 Problem 5: Takeaway

For the negative entropy regularizer

$$R(x) = \sum_i x_i \log x_i$$

on the simplex Δ_n :

Theorem (Summary)

Divergence.

$$D_{R(x,y)} = \text{KL}(x \parallel y) = \sum_i x_i \log \left(\frac{x_i}{y_i} \right).$$

Diameter.

$$\max_{x \in \Delta_n} R(x) - \min_{x \in \Delta_n} R(x) = \log n.$$

Projection.

$$\Pi_{\Delta_n}^{R(y)} = \frac{y}{\|y\|_1}.$$

Interpretation. In negative-entropy geometry, projecting a positive vector onto the simplex is not Euclidean projection. It is just normalization: divide by its ℓ_1 norm to return to a probability distribution.